

AI-Powered Investor Education Website with Conversational RAG Assistant

Project Type: Enterprise AI Web Application | Investor Education | Conversational AI

Live Platform: <https://bandhanmutual.com/iap/investment-mistakes/#rightScheme>

Project Overview

To improve investor education and engagement, working at Investigen, I developed an AI-powered web platform for **Bandhan Mutual Fund** that helps users understand common investment biases through interactive educational content and a conversational AI assistant. The objective was to replace static learning with an intelligent experience where investors could ask natural language questions and receive personalized, easy-to-understand explanations grounded in official educational content.

The platform educates users about behavioral biases such as **Herd Mentality, Loss Aversion, Confirmation Bias, Overconfidence, Recency Bias, Anchoring, Present Bias, Mental Accounting**, and other common investing mistakes, while also providing SIP/Lumpsum calculators, FAQs, educational videos, and mutual fund guidance.

My Contribution

I designed and developed the complete solution end-to-end, including:

- Developed the responsive investor education website using **HTML, CSS, JavaScript, Python, and Flask**
 - Built the conversational AI chatbot using **Google Gemini API**
 - Implemented a **Retrieval-Augmented Generation (RAG)** pipeline with **Pinecone Vector Database** for semantic search and grounded responses
 - Created the document pre-processing, embedding, retrieval, and prompt engineering pipeline
 - Designed secure **Flask REST APIs** with **request validation, API encryption, and rate limiting** for production deployment
 - Integrated SIP and Lumpsum investment calculators with the educational workflow
 - Optimized backend architecture for scalable and low-latency AI responses
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Key Engineering Decisions

Instead of relying solely on an LLM, I implemented a **RAG architecture** where every user query first retrieves relevant investor education content from a **Pinecone Vector Database** before being passed to **Gemini**, ensuring accurate, context-aware, and trustworthy responses

while minimizing hallucinations. The backend was built as modular Flask APIs with production-ready security practices, enabling scalable deployment and future feature expansion.

Technology Stack

Python • Flask • JavaScript • HTML • CSS • Google Gemini API • Pinecone Vector Database • Retrieval-Augmented Generation (RAG) • REST APIs • Prompt Engineering • API Encryption • Rate Limiting

Impact

The platform transformed traditional investor education into an interactive AI-powered experience, making complex behavioral finance concepts easier to understand through conversational guidance. By combining educational content with semantic retrieval and LLM reasoning, it increased user engagement, improved accessibility of financial knowledge, and encouraged more informed investment decisions while supporting Bandhan Mutual Fund's investor awareness initiative.